Probing the User Representation with a Decorrelated Persona Pool

The "vulnerability axis" reconsidered — and a scientific direction

User-Axis study — expanded-design note

Abstract

Our first study found a single dominant *User Axis* on Llama-3.3-70B whose PC1 correlated with a held-out vulnerability tag at r up to 0.86. That pool, however, was built by spreading personas along a vulnerability/competence spectrum, so the tag space was effectively one-dimensional — a dominant PC1 was partly *imposed*, not *discovered*. This note rebuilds the dataset as a **decorrelated factorial** over ten user attributes, holds the probing pipeline fixed, and asks what the model's user representation looks like when the attributes vary *independently*. The result overturns the headline: once vulnerability is decorrelated from competence and emotional load, it no longer drives PC1 ($\eta^2 = 0.005$); the dominant axes of user-representation variance are **topic, age, and competence**, while the psychological-state traits (vulnerability, trust, urgency, intent, style) are all weak ($\eta^2 \leq 0.10$). We argue that *dataset spread is the experiment* for this class of question, and lay out the direction it opens.

1 Motivation: from "is there a user axis?" to "what structures it?"

An instruction-tuned model silently infers who it is talking to and adapts. Our first study recovered that inference as a direction in activation space (the *User Axis* = PC1 of per-persona mean residual-stream vectors at layer 40), showed it correlated strongly with vulnerability, and showed it was behaviorally causal under steering. The natural next question is not *whether* such a direction exists but *what organizes it*: of all the ways users differ — expertise, age, topic, emotional state, intent, trust — which does the model actually encode as its primary axes?

That question cannot be answered by a pool whose attributes are entangled. If vulnerable users in the pool are also, by construction, less expert and more emotional, then a "vulnerability axis" and a "competence axis" and an "emotional-load axis" are the *same* direction, and PCA cannot tell them apart. **The dataset's spread *is* the experiment.** So we change only the stimulus distribution — decorrelate the attributes — and keep every downstream measurement identical.

2 A decorrelated factorial user pool

2.1 Design

We define ten user-attribute *factors* (Table 1) and sample 300 personas (289 surviving generation) so that each factor's levels are balanced and *crossed*: a balanced independent-shuffle followed by a short search that minimises the maximum pairwise Cramér's V. This adds axes the original pool never varied on its own — intent, domain, communication style, urgency, and age as a factor rather than a correlate of vulnerability. All persona data (backstories, elicitations, tags) is generated by Claude Sonnet 4.5, one persona per design cell.

Table 1: The ten decorrelated user-attribute factors.

factor	levels
competence	novice · intermediate · expert
vulnerability	low · moderate · high
emotional load	calm · moderate · distressed
urgency	none · moderate · acute
trust	skeptical · neutral · deferential
intent	learn · decide · vent · create · verify
domain	health · finance · legal · coding · relationships · everyday
communication style	terse-technical · plain · verbose-casual
age	teen · young-adult · adult · older-adult
on behalf of	self · caregiver

2.2 Dataset spread: how decorrelated is it?

The design achieves **max pairwise Cramér's V** $= 0.10$ (mean 0.02) across the ten factors — i.e. the attributes vary essentially independently (Fig. 1, left; contrast the ± 0.6–0.8 tag correlations of the lean-seeded pool, where expertise$\leftrightarrow$vulnerability $= -0.60$ and vulnerability$\leftrightarrow$emotional-load $= +0.77$). The decorrelation propagates to the observable judge-tags: the tag-space PCA spectrum *flattens* — PC1's share falls from 0.60 to 0.50 and PC2 rises from 0.17 to 0.25 (Fig. 1, right). The residual vulnerability$\leftrightarrow$emotional-load coupling that remains is a property of those two *scales* being near-synonymous to any annotator, not of the pool — which is exactly why we regress against the decorrelated *factors*, not the five legacy tags.

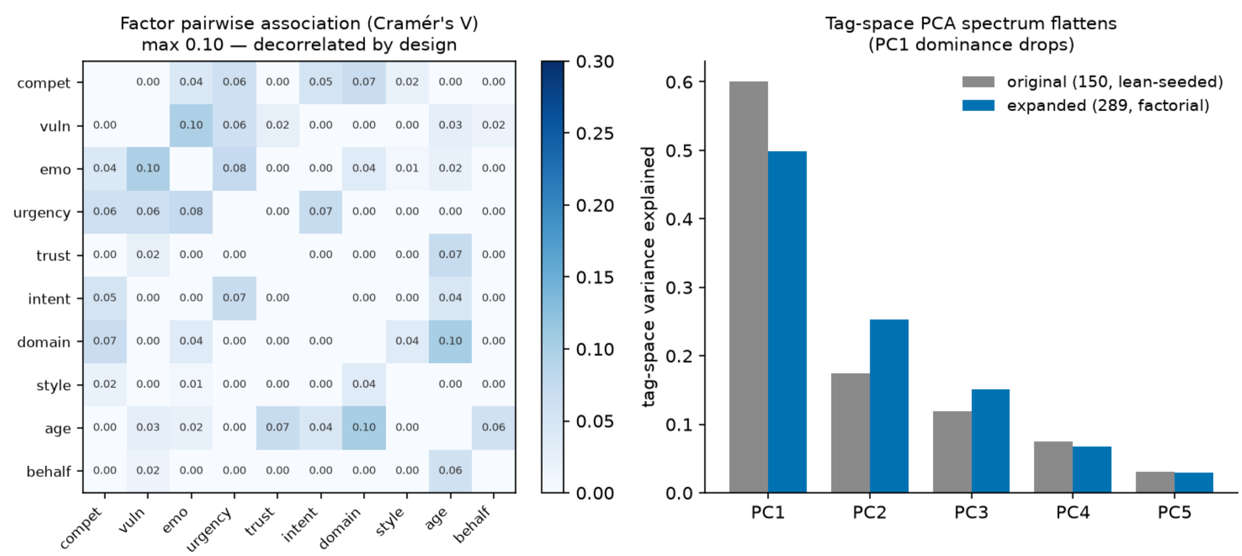

Figure 1: **Dataset spread.** *Left:* pairwise association (Cramér's V) between the ten factors over 289 personas — all off-diagonal ≤ 0.10. *Right:* tag-space PCA variance, original (150, lean-seeded) vs. expanded (289, factorial); the spectrum flattens, i.e. the pool is no longer effectively one-dimensional.

The marginals are near-uniform: each factor level ("box") holds close to $N/(\#\text{ levels})$ personas (Table 2). The thinnest boxes are the six domain levels at ~ 48 each. (Note this is a decorrelated

Latin-hypercube-style sample, not a full factorial: the *joint* cells — a specific combination of all ten factors — are almost all empty, which is what keeps the factors independent; the analysis is therefore powered at the per-factor main-effect level.)

Table 2: Marginal balance of the 289 personas — how many fall in each factor level.

factor	# lvl	personas per level
competence	3	novice 99 · intermediate 97 · expert 93
vulnerability	3	low 97 · moderate 96 · high 96
emotional load	3	calm 97 · moderate 97 · distressed 95
urgency	3	none 98 · moderate 95 · acute 96
trust	3	skeptical 98 · neutral 98 · deferential 93
intent	5	learn 58 · decide 58 · vent 57 · create 56 · verify 60
domain	6	health 48 · finance 48 · legal 49 · coding 48 · relationships 48 · everyday 48
comm. style	3	terse-technical 94 · plain 97 · verbose-casual 98
age	4	teen 71 · young-adult 70 · adult 74 · older-adult 74
on behalf of	2	self 142 · caregiver 147

2.3 Decorrelation in practice: the high-vulnerability slice

The cleanest way to *see* the decoupling is to hold one factor fixed and look at the rest. Take the 96 personas at **vulnerability** = **high**. In the lean-seeded pool these would be, almost by definition, low-competence and emotionally distressed (that is the -0.60 / $+0.77$ coupling). Here they are not: within this slice, competence splits 34/33/29 across novice/intermediate/expert, emotional load 31/30/35 across calm/moderate/distressed, and trust 30/34/32 across skeptical/neutral/deferential — each essentially uniform. A high-vulnerability user in this pool is as likely to be a *calm expert* as a *distressed novice* (Table 3). It is precisely this independence that lets PCA attribute variance to vulnerability *as such* — and find that it explains almost none ($\eta^2 = 0.005$).

Table 3: Twelve of the 96 **vulnerability** = **high** personas. Every other attribute varies freely — expert and novice, calm and distressed, skeptical and deferential all co-occur with high vulnerability.

name	competence	emo. load	trust	domain	age
Dorothy	expert	distressed	skeptical	coding	older-adult
Jasmine	expert	calm	deferential	legal	young-adult
Priya	expert	moderate	deferential	finance	young-adult
Dmitri	intermediate	calm	deferential	coding	teen
Mateo	intermediate	distressed	skeptical	finance	teen
Derek	intermediate	moderate	deferential	finance	adult
Darius	novice	calm	skeptical	finance	teen
Derrick	novice	distressed	deferential	legal	adult
Russell	novice	moderate	neutral	legal	older-adult
Phyllis	expert	distressed	neutral	relationships	older-adult
Zara	intermediate	calm	neutral	everyday	young-adult
Devin	novice	moderate	skeptical	finance	teen

The probing pipeline is unchanged from the first study: the same dual elicitation (5 explicit system-prompt phrasings, 10 in-voice openers), the same 24 shared persona-agnostic probes, the same `resp_mean/last_user` residual-stream readouts at all 80 layers, the same verbalized-inference filter

(Stage D keeps 95.9% here, MAD 1.20, even cleaner than the original 94.0%), and PCA at $\ell = 40$. Holding the measurement fixed makes any change in the discovered axis structure attributable to the decorrelated design.

3 What the balanced pool reveals

We regress each principal component's per-persona loading on each factor, reporting η^2 (fraction of the PC's spread explained by that factor; one-way ANOVA, BH-FDR).

The vulnerability axis dissolves. PC1's correlation with the vulnerability *tag* drops from 0.78 to 0.54 (`resp_mean`) and from 0.86 to 0.55 (`last_user`), and PC1's variance share falls from 0.20 to 0.17. Decisively, PC1's η^2 for the *independent* vulnerability factor is **0.005** — essentially zero. What PC1 actually tracks at $\ell = 40$ is **competence ($\eta^2 = 0.17$), emotional load (0.13), and topic (0.13)**. In other words, the original "PC1 = vulnerability" finding was inflated by vulnerability being entangled with competence and emotional load; the pure, decorrelated vulnerability signal is not a dominant direction.

The dominant axes are topic, age, and competence. Across PC1–PC5 (Fig. 2), the strong movers are **domain** (η^2 up to 0.64 on the `last_user` readout), **age** (up to 0.45), and **competence** (up to 0.36). Crucially, domain dominates even on the *pre-response* `last_user` token — so it is the model organising its user representation by topic, not merely response content leaking into `resp_mean`.

The psychological-state traits are weakly represented. Vulnerability, trust, urgency, intent, communication style, and self-vs-caregiver are all weak ($\eta^2 \leq 0.10$) on every one of the top five components. Whatever the model encodes most strongly about its user, in a balanced pool, is *who they are* (topic, age, expertise) far more than *what state they are in.*

4 Comparison to the lean-seeded pool

Table 4 makes the reversal explicit. The two studies use the same model, layer, probes, and pipeline; they differ only in how the persona pool is spread. That single change moves PC1 from "the vulnerability axis" to "a competence/topic axis on which the independent vulnerability manipulation has no measurable effect."

This is not a contradiction of the first study so much as a correction of its interpretation. The axis is real and reproducible; what was over-claimed was its *identity.* A direction that separates the personas we *built* to differ in vulnerability will of course correlate with vulnerability — but that is a statement about the pool, not about the model's priorities.

5 This as a scientific direction

The broader lesson generalises beyond vulnerability: **when you read a latent attribute off a model by contrasting hand-authored stimuli, the stimulus distribution can manufacture the very axis you report.** De-correlating the design is the antidote, and it turns "does

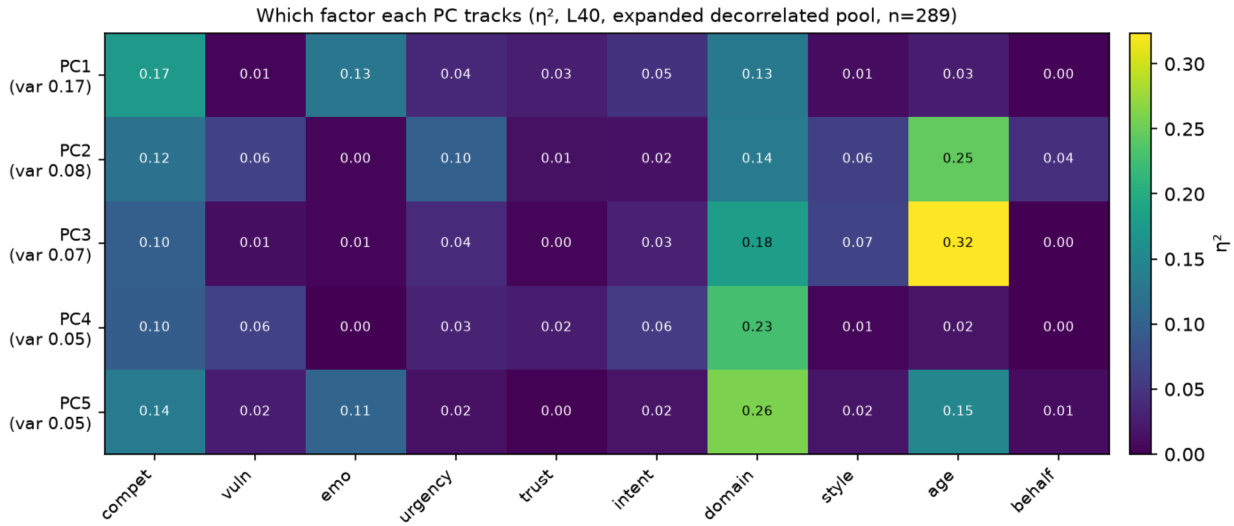

Figure 2: **Which factor each PC tracks** (η^2, `resp_mean`, $\ell = 40$, $n = 289$). PC1 loads on competence/emotional-load/topic — *not* vulnerability ($\eta^2 = 0.01$). The bright columns are domain and age; the vulnerability, trust, urgency, intent, style, and on-behalf columns are dark throughout.

Table 4: Original (lean-seeded, $n = 150$) vs. expanded (decorrelated factorial, $n = 289$); `resp_mean` at $\ell = 40$.

	original	expanded
pool design	lean-seeded (22 archetypes)	decorrelated factorial (10 factors)
attribute entanglement	bundled ($\|r\|$ up to 0.77)	decorrelated (V $\leq$ 0.10)
tag-space PC1 share	0.60	0.50
PC1 activation variance	0.204	0.173
PC1 $\sim$ vulnerability tag	**0.78**	0.54
PC1 η^2, vulnerability *factor*	— (confounded)	**0.005**
PC1 top factors	(vulnerability bundle)	competence, emo-load, topic
dominant axes overall	vulnerability	**topic, age, competence**

the model represent X?" into a sharper, falsifiable question: *how much of the representation's variance does an independently-manipulated X explain, holding everything else fixed?* Under that test, several attributes the field treats as central to "user modelling" (vulnerability, trust, intent) turn out to be weakly encoded, while topic and age dominate.

We see three concrete threads:

- **Disentangle topic from user-state.** Domain has the most levels and the largest η^2; partialling it out (or matching domain across the state factors) would sharpen how much genuine *state* structure remains. That the signal is strong even on the pre-response token suggests the model's "user model" is substantially a "conversation-topic model."
- **Blind, held-out axis naming on the balanced pool.** Re-running the unsupervised auto-labeling (a judge names the extreme poles of each PC, validated on held-out personas) on this pool would give an interpretation-free account of what topic/age/competence axes actually separate.

- **Causality per axis.** Steering along the new competence/age/topic directions — and testing whether steering moves the model's *verbalized* user model, not just its register — would establish which of these representations are behaviourally load-bearing rather than merely present.

Caveats. One model, one layer family; `resp_mean` is response-content-laden (mitigated but not eliminated by the `last_user` check); and η^2 favours factors with more levels (domain has six). None of these overturn the central result: the dominant, decorrelated axes of the model's user representation are demographic and topical, and the vulnerability axis was, to a substantial degree, a property of the pool we built.

A Intrinsic vs. extrinsic user attributes

The ten factors — and the columns of Table 3 — fall into two kinds. **Intrinsic** attributes are stable properties of the *person* that travel across conversations (who they are); **extrinsic** attributes are properties of the *current situation* — the topic on the table, the user's transient state, their stance toward the assistant, and their goal for this turn (what is going on right now). Table 5 classifies each factor.

Table 5: Intrinsic (stable person-traits) vs. extrinsic (situational) attributes. The columns of Table 3 that are stable traits (competence, age) sit above the line; the situational ones (emotional load, trust, domain) sit below — and within the **vulnerability** = **high** slice both kinds vary freely.

factor	class	what it is
competence	intrinsic	subject-matter skill; stable across conversations
age	intrinsic	demographic; fixed
communication style	intrinsic	habitual register / verbosity
domain	extrinsic (context)	the *topic* of this conversation, not a person-trait
vulnerability	extrinsic (state)	susceptibility to harm in this situation
emotional load	extrinsic (state)	how distressed the user is *right now*
urgency	extrinsic (state)	time pressure of the current task
trust	extrinsic (stance)	attitude toward the assistant in this interaction
intent	extrinsic (goal)	what the user wants from this turn
on behalf of	extrinsic (relational)	self vs. asking for someone else

This split sharpens the main result. The axes the model encodes most strongly are the two *intrinsic* traits (age, competence) plus the *extrinsic context* (domain/topic); the *extrinsic state* attributes — vulnerability, emotional load, urgency, trust, intent — are all weak ($\eta^2 \leq 0.10$). In short, the model's user representation captures **who the user is** and **what the conversation is about** far more than **what state the user is in**. That is a natural thing for a next-token predictor to encode — stable identity and topic are strongly predictive of the surface form of the exchange, whereas fine-grained emotional or motivational state is not — but it is precisely the state dimension that safety arguments about "vulnerable users" rely on.